

CHOLESTEROL OXIDATION ALTERS CRITICAL ALLOSTERIC INTERACTIONS BETWEEN GLYCINE TRANSPORTER 2 AND MEMBRANE CHOLESTEROL

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ABSTRACT

Neurotransmitter transporters such as Glycine transporter 2 (GlyT2) are important pharmaceutical targets that are exquisitely sensitive to membrane cholesterol. Membrane cholesterol binds to GlyT2 at the membrane embedded CHOL1 site, modulating conformational cycling, substrate uptake, and the affinity of analgesic allosteric inhibitors.^{1,2} Buildup of oxidized cholesterol species is a hallmark of neurodegenerative disease, altering membrane biophysical properties.³ Using coarse-grained molecular dynamics, we investigate how membrane oxysterol content perturbs allosteric cholesterol binding to GlyT2. We find oxysterols profoundly alter GlyT2/lipid interactions in an oxidation-site specific manner. Oxysterols are strongly enriched in the GlyT2 lipid annulus, displacing cholesterol and polyunsaturated lipids. Furthermore, tail-oxidised cholesterol directly outcompetes cholesterol at the CHOL1 site through exceptionally stable bidentate binding, which impairs cholesterol-sensitive GlyT2 inhibitors. Our findings reveal how oxysterol accumulation may dysregulate transporter function and alter therapeutic responses in disease-altered neuronal membranes.

REFERENCES

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